

Diamond tools in stone and civil engineering industry: cutting principles, wear and applications

H.K. Tönshoff, H. Hillmann-Apmann*, J. Asche

Institute for Production Engineering and Machine Tools, University of Hannover, Schlosswender Strasse 5, D-30159 Hannover, Germany

Abstract

Metal-bonded diamond tools have been introduced to the processing of natural stone and in civil engineering applications several decades ago. They have, without any doubt, revolutionised the whole sector by allowing dramatically increased rates of material removal. However, although significant improvements in tool performance have been made since the early days, diamond tools are still the main factor for cost effectiveness. For this reason, it is of prime importance to possess detailed knowledge about the cutting principles and wear performance. Only from these basics is it possible to find optimum process parameters and to evaluate the effects on the diamonds during a given process. Thus, the prediction of suitable processing conditions can be achieved. This paper gives an overview of process principles and their relevance for tool and process design. With respect to applications in the stone and civil engineering industries, some trends will be outlined. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Stone cutting; Diamond tools; Diamond wear; Stone processing

1. Introduction

Diamond tools govern cutting processes in the natural stone industry and in similar fields of application. One of the most widespread processes is the so-called ‘sawing’ that is applied not only in the natural stone field, but also in civil engineering. For this reason, all the following sections refer to conditions when using circular tools. However, all the main aspects also hold true for other processes in this field when diamonds are involved.

2. Cutting principles

Chip formation can be described as destruction of a material consistency by utilising a tool. This is common for all materials. In cutting stone, some particular features must be taken into account. First, it must be stated that the process is a grinding process, although in general the term ‘sawing’ is commonly used. Models based on simple geometries of cutting edges divide the process

of formation of a cut into two mechanisms, which are shown in Fig. 1.

Factors influencing this process are directly or indirectly contained in this model:

1. Physical material properties of the stone, such as grain size and strength;
2. Forces between the diamonds and the material;
3. Stress distribution in the rock; and
4. Temperatures in the tool–workpiece interface.

When cutting stone with disc-like diamond tools, the mechanical interaction of tool and workpiece results in process forces, mainly caused by the following factors:

- Elastic and plastic workpiece deformation by the cutting edges;
- Friction between the stone and the diamonds;
- Friction between the stone and the matrix; and
- Friction between the swarf and the matrix.

In front of a grain that is engaged in the process, stresses are effected by tangential forces. In this zone, swarf is processed by tensile and compressive stresses. This mechanism is called primary chip formation. The swarf is forced out through grooves in front and beside the grain [1].

*Corresponding author. Tel./fax: +49-511-762-3582.

E-mail address: hillmann@ifw.uni-hannover.de (H. Hillmann-Apmann).

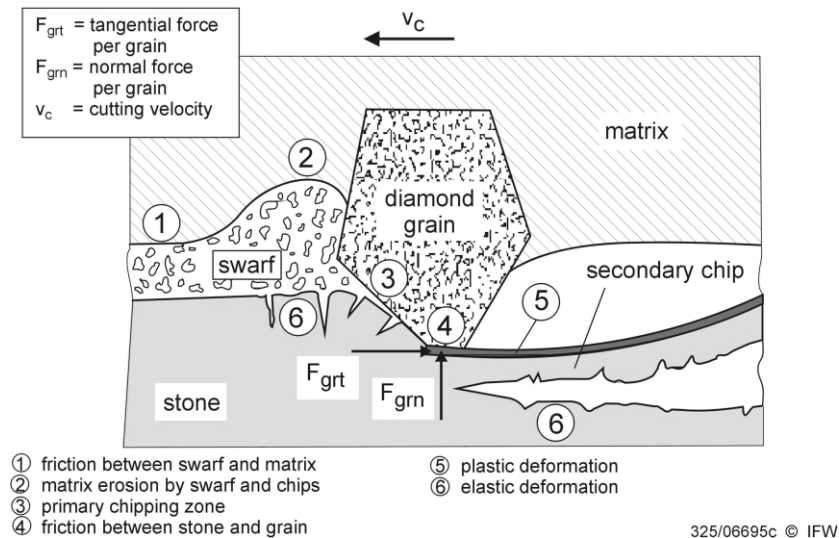


Fig. 1. Mechanical interaction between tool and workpiece when cutting stone.

While the rock shows elastic characteristics up to its ultimate stress, it is necessary for cutting to reach a certain minimum grinding thickness. The material to be cut is deformed by the compressive stress conducted below the diamond. When the load is removed, an elastic reversion leads to critical tensile stresses, which cause brittle fracture. This mechanism effected by tensile stresses is termed secondary chip formation [1]. The swarf generated as a result of the process is carried away by the coolant.

3. Tools for stone cutting

For the processing of blocks of natural stone to modular elements, such as tiles, mainly circular diamond

tools are utilised today (see Fig. 2). These tools consist of two main elements, a steel centre and a segment.

The tool design and the choice of adequate machining parameters belong to the know-how and experience of the toolmakers and the end-users, respectively.

3.1. Steel centre

The steel cores are made from different steel grades, depending on whether the segments are brazed or laser-welded. Of course the steel cores are hardened and tempered to approximately 43–45 HRC. Normally, the final form of the core is made by laser cutting or milling. The borehole is finally made by honing, because

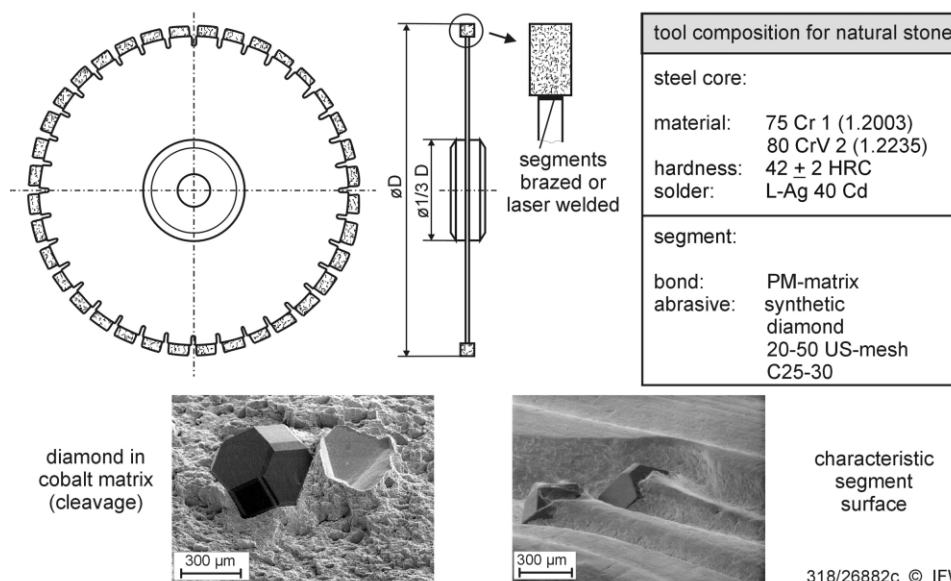


Fig. 2. Circular tools for stone cutting.

it is necessary to obtain excellent roundness with extremely small tolerances. A very important point is the exact thickness (variation of ± 0.01 mm) over the total geometric form. This is achieved by special grinding machines. In order to avoid tool run-out during cutting, residual stresses have to be brought into the blade core by thermal or mechanical treatment (tensioning). Parallel blade sets have to be similarly treated to assure constant behaviour of all blades over the whole tool life.

3.2. Segment design

The segments are attached to the steel core by brazing or laser-welding. The wear resistance of the matrix has to be adjusted to the wear speed of the diamond: if the matrix wears too fast, the diamond capacity is not completely used; if it wears more slowly than the diamonds, the space between the cutting edges and the matrix is constantly reduced. The swarf cannot be carried away properly, and the segment will continuously lose its ability to cut. Thus, a relatively soft matrix has to be used for hard stone, such as granite. In general, soft matrices also offer better diamond retention.

Cutting segment developments are focused on the one hand on the manufacturing process of the segments, and on the other hand on segment materials (diamonds and metal powders) and their properties. Production technology of diamond-impregnated cutting segments mainly consists of a hot isostatic-pressure treatment (HIP), in which a so-called green pre-moulded segment is finally pressed. Over the last few years a technology has reappeared that works by a technique of hot sintering without pressure, better known as 'free sintering'. Compared to the conventional HIP process, this technique offers some decisive advantages, even as an intermediate sintering process. With free sintering, it is possible to achieve a much closer porosity of the metal powder. Another singular aspect of free sintering is that during the volumetric reduction by cooling (shrinkage), the diamond grains are forced towards the inside of the segment. However, the movement progressively diminishes with an increase in density of the sintered piece, so that partially projecting grains can be obtained, which are ready to carry out their abrasive action without the need for grinding-in of the tool [2].

Looking at the diamond crystals and matrix materials, the latest developments tend to use high-quality diamond grits with either a titanium or chromium coating [3], together with wear-resistant bonds with a low cobalt content. Diamond coating techniques nowadays offer much better retention of the diamond crystal in the matrix material.

In order to meet the latest environmental requirements and economical needs, new metal powders to be used in a partial substitution of cobalt in diamond segment

bonds have been introduced to the market [4]. These powders have properties somewhat similar to pure cobalt. They are ultra-fine alloyed powders, composed of copper, iron and cobalt, with each elementary grain containing these basic three metals. After sintering, such powders give a very homogeneous, ultra-fine microstructure. In particular, the compatibility of these powders with diamonds and coated diamonds are proposed to offer high potential for tool performance.

4. Wear of diamond tools

Affected by the mineral composition of the stones, different segment wear-states can be found, depending on the components of the cutting segment, as well as on the machining parameters. For the above-discussed application, wear of segments can be divided into matrix wear on the one hand and diamond wear on the other hand.

One mechanism that directly affects the diamond is the abrasive effect of the stone material, together with high mechanical and thermal loads particular to the process, which lead to wear of the grains. On the other side, the coolant in combination with the swarf forms an abrasive material that causes erosion of the segment matrix. Both mechanisms have a major influence on the wear behaviour and the results of the process.

Wear caused by swarf mainly depends on the composition of the stone. Swarf consisting of particles from metamorphic and deep-seated rock shows smaller grains than the tool material. In the cutting process, sedimentary rock mainly fractures along its grain boundaries because of the low strength of its binding material [5,6]. This results in a swarf consisting of small and hard grains. In this case, the wear effect on the diamond grain is unimportant compared with the high abrasiveness of the swarf, which leads to a large wear rate of the matrix.

4.1. Diamond wear

Wear mechanisms effecting the diamond grains can be divided into four main forms:

1. Adhesion wear: the diamond sticks to the stone surface and particles are sheared off.
2. Friction wear: very hard grain particles of the rock scratch the diamond surface.
3. Wear by diffusion: chemical reactions between the workpiece and the diamond surface lower the strength or the hardness.
4. Fracture of grains: fracture of the diamonds is caused by mechanical or thermal overload or by fatigue.

Previous investigations have shown that wear by adhesion and diffusion is not significant in the processing of stone [7]. Fracture of the diamonds and wear by

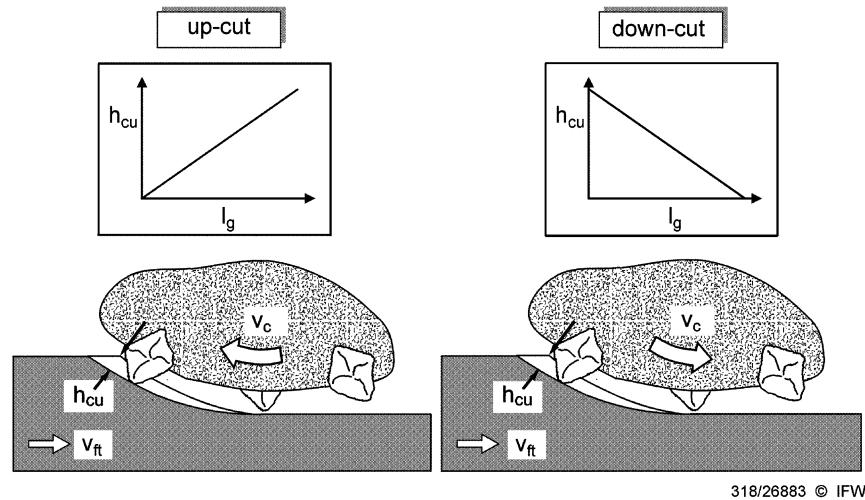


Fig. 3. Single-grain chip thickness in up- and down-cutting mode.

friction are the major wear effects that cause a reduction of the grains. The wear form depends on the different loads during the cut. Under up-cutting conditions with a low cutting speed and a homogenous stone, the diamonds will be worn flat. High feed speeds under down-cutting conditions in processing a stone with hard zones will lead to fractured diamonds because of the shock load.

A commonly used characterisation has been carried out by Schulze [7], who identifies two different wear states between a fresh and a pulled-out grain:

- Worn flat; and
- Fractured grain.

The worn-flat grain is caused by high thermal load on the grain tip, while the mechanical impulse is not sufficient for fracture, i.e. the grain face area is too small and/or the cutting velocity too low. Fracture is caused by a mechanical impulse, requiring a certain grain face area and/or cutting speed [7].

4.2. Matrix wear

Besides wear of grains, the abrasive swarf consisting of coolant and stone particles leads to an erosion of the bond. Depending on the stone being cut, the bond material has to be chosen very properly. Otherwise, a division into two cases of matrix wear can be made:

- If the matrix wears too fast, the diamond capacity is not completely used before the grain is pulled-out.
- If the matrix wears more slowly than the diamonds, the space between the cutting edges and the matrix is constantly reduced. The swarf cannot be carried away properly, and the segment will continuously lose its ability to cut.

A characteristic feature of matrix wear is a crater in front of the single grains. Behind the grain, there is less erosion of the matrix; a bond tail is formed, so that the diamond is sustained against the process forces to remain in the bond.

4.3. Influences of machining parameters and kinematics

The material removal rate is generally limited by the maximum thermal and mechanical loads that the diamond grains can withstand. The load on the diamonds is, to a great extent, influenced by the machining parameters. At a certain material removal rate per unit, it depends on the machining conditions as follows [8]:

- a_c : the greater the depth of cut, the smaller is the active grain-face area. Because of the greater contact length, the thermal load of the diamond tip is elevated.
- v_f : the higher the feed velocity, the larger is the active grain-face area, and the smaller is the thermal load on the diamond.
- v_c : the higher the grinding velocity, the higher is the thermal load on the grain tip. The active grain-face area becomes smaller, but the mechanical shock impulse at the beginning of the chip removal becomes shorter, and thus more prominent.

Besides machining parameters, a major influence can be found in the process kinematics. If the operation is carried out in down-cut mode, a single grain reaches its maximum chip thickness at the beginning of the contact path (see Fig. 3). The contrary holds true for the up-cut mode, where the chip thickness increases to its maximum at the end of the contact. In this case, mainly elastic deformation occurs at the beginning, and just when a critical chip thickness is reached, chips are formed [1]. This leads to different wear characteristics

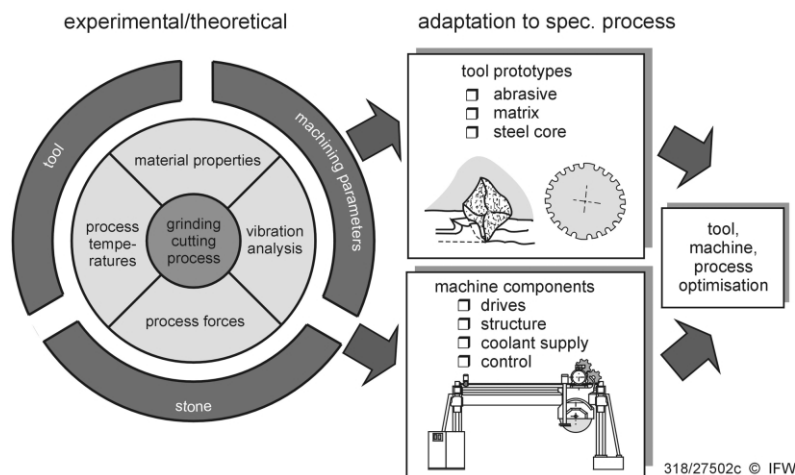


Fig. 4. Analysis and optimisation of stone cutting processes.

in up- or down-cut modes. In the down-cut process, high mechanical-shock impulses are acting, depending on the type of abrasive and bond, and diamond fracture and/or pull-out are likely. In up-cut processes, the chip thickness is increased from zero to maximum and only the tip of the diamond is first in contact with the stone material, resulting in friction and flattened grains.

5. Process design and optimisation techniques

In a variety of cases, cutting processes do not achieve their optimum performance, or even show unsatisfactory results. In these cases, inadequate tool, machine and process designs must be taken into account and a systematic approach has to be considered, in which investigations of all the relevant data for this process have to be considered. Measurement and analysis of material properties, process forces, temperatures and vibrations have to be carried out to gain basic knowledge as a starting point for improvements in the tool and machine (see Fig. 4). In particular, mechanical and thermal loads at the tool–workpiece interface have to be viewed in detail. These factors not only depend on machining parameters, but also to the same degree on the material to be cut and vibrations occurring in the process, and thus a complete evaluation of the single factors and their interrelations often has to be carried out.

Having successfully completed basic investigations with the identification of weak points, improvements can begin. These have to fulfil the needs of both tool and machine. To obtain knowledge of diamond tool wear, investigations on the progression of matrix and diamond wear must be carried out. In order to find optimum process parameters and tool components, it is obvious that knowledge about the wear process must be obtained. This comprises effects on the diamonds and on the bond during the cut. The background for all

these investigations is, of course, the economic point of view of reducing the significant costs for these expensive tools. In general, tools have to fulfil the main characteristics of having high vibrational stability, defined vibration behaviour of the blade and the best possible wear resistance.

The technical approach to these goals needs to include, besides theoretical analysis, a series of cutting experiments, additionally aimed at optimised machining parameters. Both aims have to lead to minimum vibrations of tool and machine for high workpiece accuracy, as well as a low noise level, optimum productivity and minimum tool wear rate.

These goals can be achieved on the one hand by evaluation of the machine and tool vibration behaviour by means of modal analysis. The results will help to avoid cutting at critical speeds of the structure that result in elevated vibration amplitudes. On the other hand, the only variable for increased productivity is the feed speed, which has to be as high as possible. The grinding velocity can be utilised to increase the dynamic stiffness of the blade and to shift between thermal and mechanical load of the grains. The higher the grinding velocity, the higher is the dynamic stiffness of the blade and the thermal load on the diamond grains, but the lower is the active grain-face area and the tangential force on the grains. An appropriate balance has to be found. This can be achieved by force and wear measurements under different machining conditions.

In addition, machine components, such as the machine structure, spindle drives, girders and bearings, have to be checked and improved or adapted if necessary.

6. Summary and outlook

The application of diamond tools, in this case for cutting of natural stone, can be described as a complex system with a large variety of influencing factors.

However, as explained in the previous sections, taking into account the basic process (and it must be emphasised that it is a grinding process) and all relevant characteristics, solutions can be found for most problems that arise in cutting applications. In most cases, only this systematic approach offers the possibility of detecting weak points, which are not obvious and not solvable when regarding single aspects alone.

Besides the technological point of view that has been discussed in this paper, socio-economic and socio-ecological aspects have to be considered as well. This especially holds true for highly industrialised countries, where not only technology, but also the significance of man and environment is increasingly the focus of attention. Regarding, e.g. the policies of the European Community, it is obvious that support for competitiveness and sustainable development will only be given in the future if these can simultaneously create wealth and employment, enhance the quality of life of citizens and preserve the environment and natural resources. This must, of course, proceed while ensuring the competitiveness of industry. For this reason, approaches have to consider that new knowledge, advanced technologies and innovative solutions must be generated by companies and public authorities alike, resulting in high-quality, innovative goods, processes and services. This can be pursued by a systematic approach, where research supports the development of coherent, interconnected and eco-efficient industries capable of responding to both the market and the general needs of society.

Depending on these overall conditions, future trends and potential strategies for R&D can be divided into technology and socio-economic subjects. This reflects the above policies, but also shows that there are obviously interrelations between the single subjects. The one side does not necessarily exclude or have a negative influence on the other side. Thus, environmental requirements also have an economic value like, e.g. waste minimisation, where a higher output can be achieved when the cutting kerf width is decreased.

References

- [1] W. Ertingshausen, Zerspanung von Granit mit Diamant-Trennschleifscheiben, Universität Hannover, 1984, Dr-Ing dissertation.
- [2] O. Cai, Sintering without pressure or free sintering, *Diamante* 11 (1997) S106–122, Milan.
- [3] G. Bräuninger, S.C. Hayden, MBS*-960 CR2 and MBS-960 Ti2, *Diamante* 7 (1996) S73–83, Milan.
- [4] M. Bonneau, Mechanical properties of special ternary alloy bonds compared with cobalt bonds, Conference on PM Diamond Tools, Montreux, 1997.
- [5] G. Mehling, *Natursteinlexikon für Handwerk und Industrie*, Callwey Verlag, München, 1986.
- [6] H.K. Tönshoff, H.-G. Wobker, R. Przywara, Das Arbeitsverhalten von Werkzeugen zum Trennschleifen von Gestein, *Ind. Diamanten Rundschau* 3/93 (1993) S198–205.
- [7] R. Schulze, *Trennschleifen von Hartgestein*, Universität Hannover, 1980, Dr-Ing Dissertation.
- [8] H.K. Tönshoff, J. Asche, Verschleißverhalten von Diamantwerkzeugen in der Natursteinbearbeitung, *Ind. Diamanten Rundschau* 2/96 (1996) S107–114.